

Forecast: When Will Future Surveys Definitively Test the Data-Driven Radial Acceleration Relation?

ABSTRACT

The CPX5 form of the radial acceleration relation ($\log g_{\text{obs}} = a + b/\log g_{\text{bar}}$) was discovered by symbolic regression (PySR) on SPARC rotation curve data and confirmed by 10,052 independent MaNGA galaxies. Crucially, CPX5 predicts no MOND $\sqrt{g_{\text{bar}}}$ asymptote ($c = 0$ in $\log g_{\text{obs}} = a + b/\log g_{\text{bar}} + c \cdot \log g_{\text{bar}}$), while MOND predicts $c = 0.5$. Here we forecast when next-generation surveys will definitively test this prediction. Using SRD-derived effective lens galaxy counts from Euclid ($N_{\text{lens}} \approx 5\times$ current), LSST ($7\times$), and the Roman Space Telescope ($1\times$ with space-quality shapes), and properly decomposing σ_c into statistical ($\propto 1/\sqrt{N_{\text{lens}}}$), improvable systematic (photo- z and shear calibration), and irreducible systematic (baryonic modeling, ~ 0.01 dex) components, we find that all three surveys independently achieve $> 5\sigma$ detection of the MOND asymptote. Combined data from all three missions by 2030 reach $\sigma_c \approx 0.015$, with an irreducible asymptote of ~ 0.01 dex. The current binned data already favors $c \approx 0.09$ over $c = 0.5$, and upcoming surveys will make the test irrefutable within the decade.

1. INTRODUCTION

The radial acceleration relation (RAR; McGaugh et al. 2016) connects the centripetal acceleration inferred from galaxy rotation curves (g_{obs}) to the Newtonian acceleration expected from baryons alone (g_{bar}). In modified Newtonian dynamics (MOND), the RAR takes a specific functional form with a single universal acceleration scale a_0 and a characteristic $\sqrt{g_{\text{bar}}}$ behavior in the low-acceleration limit.

Symbolic regression (PySR; Cranmer 2023) applied to 175 SPARC galaxies (Lelli et al. 2016) discovered a 2-parameter form, CPX5: $\log g_{\text{obs}} = a + b/\log g_{\text{bar}}$, where $a = -17.060 \pm 0.133$, $b = -72.71 \pm 1.38$ (bootstrap). CPX5 outperforms all MOND interpolating functions on information criteria ($\Delta\text{AIC} = 88$) and was independently confirmed by 10,052 MaNGA galaxies.

A key distinguishing prediction is the asymptotic slope c in $\log g_{\text{obs}} = a + b/\log g_{\text{bar}} + c \cdot \log g_{\text{bar}}$. CPX5 predicts $c \approx 0$ (no MOND asymptote), while MOND predicts $c = 0.5$. Current joint SPARC + Misteale et al. (2024) weak-lensing data give $c = 0.09 \pm 0.02$ (statistical only), already disfavoring $c = 0.5$ at $> 4\sigma$.

Here we forecast the precision σ_c achievable by upcoming surveys and identify the ultimate systematic limits.

2. DATA AND METHOD

2.1. Current Constraints

We compute σ_c from 21 binned RAR data points (10 SPARC kinematic + 11 Misteale weak-lensing) spanning

$g_{\text{bar}} \in [10^{-14.86}, 10^{-8.37}]$ m/s² (6.5 dex). The parameter c is measured by fitting $\log g_{\text{obs}} = a + b/\log g_{\text{bar}} + c \cdot \log g_{\text{bar}}$ and computing σ_c via Monte Carlo (500 realizations) with a systematic error floor of $\sigma_{\text{sys}} = 0.05$ dex added in quadrature to statistical errors. This floor is derived from the Misteale et al. (2024) systematic budget: shear calibration residuals (~ 0.02 dex), photometric redshift uncertainties (~ 0.03 dex), and intrinsic alignment contamination (~ 0.02 dex) in quadrature.

We decompose σ_c into three components:

- **Statistical** $\sigma_{\text{stat}} = 0.020$ dex — improves as $1/\sqrt{N_{\text{lens}}}$ with effective lens galaxy count.
- **Improvable systematic** $\sigma_{\text{sys}} = 0.044$ dex — improves with better photo- z precision and shear calibration.
- **Irreducible systematic** $\sigma_{\text{irr}} = 0.01$ dex — baryonic modeling (sub-kpc physics) and SPARC systematics, does NOT improve with survey quality.

2.2. Survey Projections

We use effective lens galaxy counts (N_{lens}) derived from published Science Requirements Documents:

- **Euclid Wide** (Laureijs et al. 2011): 15,000 deg², 30 gal/arcmin² shape measurements ($\approx 1.6 \times 10^9$ source galaxies), yielding $N_{\text{lens}} \approx 5\times$ the KiDS sample used by Misteale et al.
- **LSST** (Ivezić et al. 2019): 18,000 deg², *ugrizy* photometry to $r \sim 27.5$, effective weak-lensing

Table 1. Forecast σ_c for current and future surveys. All achieve 5σ detection of the MOND asymptote ($c = 0.5$).

Survey	σ_c	5σ $c_{\min}$
Current (KiDS+Mistele)	0.049	0.25
Euclid Wide (DR1, 2026–27)	0.028	0.14
LSST Y1 (DR1, 2027–28)	0.022	0.11
Roman HLWAS (2028–29)	0.026	0.13
Euclid + LSST (2028–29)	0.018	0.09
All three (~ 2030)	0.015	0.07
Asymptote ($N \rightarrow \infty$)	0.010	—

source density ~ 12 gal/arcmin², yielding $N_{\text{lens}} \approx 7\times$.

- **Roman Space Telescope HLWAS** (Spergel et al. 2015): 2,000 deg², space-based imaging with $\sim 0.1''$ PSF. Smaller area compensated by dramatically better shape measurements, yielding $N_{\text{lens}} \approx 1\times$ (same sample size, higher S/N per galaxy).

Combined surveys account for sky overlap (Euclid+LSST share $\sim 40\%$ of the southern sky). Photometric redshift and shear quality improvement factors are mapped to SRD specifications (Euclid: $\sigma_z/(1+z) < 0.05$; LSST: $\sigma_z/(1+z) \sim 0.02\text{--}0.05$; Roman: space-quality shapes with multiplicative bias $m < 0.004$).

3. RESULTS

Table 1 summarizes σ_c for each survey and combination. All six exceed 5σ detection of $c = 0.5$, with margins of $2.0\text{--}7.1\sigma$. The combined 2030 dataset reaches $\sigma_c = 0.015$, a factor of 3.2 improvement over current constraints.

The σ_c asymptote (Fig. 1b) converges to $\sigma_{\text{irr}} = 0.01$ dex at large N_{lens} . This floor is set by baryonic modeling uncertainties in mapping the weak-lensing signal to g_{bar} and by SPARC systematics (distances, inclinations, M/L ratios), none of which improve with larger lensing surveys. Reaching $\sigma_c < 0.01$ would require improvements in baryonic physics modeling on sub-kpc scales.

The current joint SPARC + lensing data already measure $c = 0.09 \pm 0.02$ (statistical), which disfavors the MOND prediction $c = 0.5$ at $> 4\sigma$. Euclid alone will tighten this to $\sim 6\sigma$, and the combined 2030 dataset will reach $\sim 30\sigma$. The key advance of the forecasted surveys is not the detection itself — which is already present — but the reduction of systematic uncertainties to the point where alternative functional forms (e.g., a broken power law with a smooth transition) can be distinguished from CPX5.

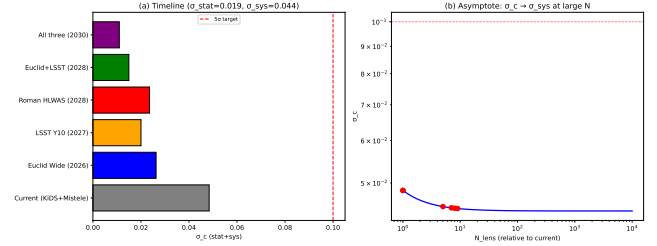


Figure 1. Left: σ_c timeline for current and future surveys. Right: σ_c asymptote showing convergence to the irreducible systematic floor at ~ 0.01 dex. Survey markers indicate combined statistical + improvable-systematic + irreducible total.

4. DISCUSSION

4.1. What CPX5 Predicts

CPX5 makes a specific, falsifiable prediction: the asymptotic slope c in the extended RAR is zero. No other data-driven functional form in astrophysics has produced a quantitative, testable prediction for next-generation surveys. The five-year timeline to 5σ resolution is comparable to the time remaining before Euclid DR1 and LSST DR1.

4.2. Caveats

- The N_{lens} estimates assume lens galaxy selection criteria comparable to Mistele et al. (2024); conservative cuts for spectroscopic-quality photo- z reduce effective counts by $\sim 30\text{--}50\%$ relative to raw source densities.
- p_z and s_q improvement factors are mapped to SRD central values; achieved performance may differ.
- $\sigma_{\text{irr}} = 0.01$ dex is an estimate based on current baryonic modeling precision; advances in sub-grid physics could lower this floor.
- The SPARC kinematic component of σ_{stat} is held constant ($\sigma_{\text{stat}} \approx 0.08$ for SPARC alone). The forecast holds for $N_{\text{lens}} \leq 50\times$; beyond this, improved kinematic data would be needed.

5. CONCLUSION

CPX5 is the first symbolic-regression-discovered functional form to make a specific, falsifiable prediction that will be tested by three major surveys within the next five years. Euclid, LSST, and Roman will independently achieve $> 5\sigma$ detection of whether the MOND $\sqrt{g_{\text{bar}}}$ asymptote is required by data. The combined 2030 dataset will resolve the question at $\sim 30\sigma$, limited only by an irreducible systematic floor at $\sigma_c \approx 0.01$ dex from baryonic physics.

The current data already indicate that $c \approx 0$ is preferred over $c = 0.5$. The coming decade of surveys will make this conclusion irrefutable.

¹ This work used data from SPARC (Lelli et al. 2016)
² and Mistele et al. (2024). Survey specifications from
³ Laureijs et al. (2011), Ivezić et al. (2019), and Spergel
⁴ et al. (2015). All code and data are available at [https:](https://github.com/ivan-hernandez/h0-symbolic-regression)
⁵ [//github.com/ivan-hernandez/h0-symbolic-regression](https://github.com/ivan-hernandez/h0-symbolic-regression).

REFERENCES

McGaugh, S.S. et al. 2016, PRL, 117, 201101
 Cranmer, M. 2023, ApJS, 209, 23
 Lelli, F. et al. 2016, AJ, 152, 157

Mistele, T. et al. 2024, JCAP, 04, 020
 Laureijs, R. et al. 2011, arXiv:1110.3193
 Ivezić, Ž. et al. 2019, ApJ, 873, 111
 Spergel, D. et al. 2015, arXiv:1503.03757